Epileptic Seizure Prediction Using EEG Data: Investigating Preprocessing, Regularization, and Class Imbalance Handling

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***Abstract*—This paper presents a comprehensive empirical study of epileptic seizure prediction using three real-world EEG datasets with distinct characteristics. The study investigates how preprocessing pipeline design (SelectKBest feature selection versus PCA dimensionality reduction), regularization strategy (L1/Lasso, L2/Ridge, Elastic Net), and class imbalance handling (SMOTE, random undersampling, class weighting) interact to affect model generalization performance. A Logistic Regression classifier is trained across all configurations using an 80/20 stratified split. Results show that Pipeline A (SelectKBest) outperforms Pipeline B (PCA) by 18.25 F1 points on the clinical dataset (DS2). On the highly imbalanced UCI EEG dataset (DS3, 1:4 ratio), SMOTE raises seizure Recall from 0.037 to 0.461, a $12\times$ improvement, while class weighting provides comparable gains. All three regularizers achieve identical perfect scores on DS1 and DS2; Elastic Net achieves the highest F1 on DS3 with SMOTE (F1$=0.342$). L1 exhibits extreme sparsity — 63.6% on DS1 and 95% on DS2 — confirming its implicit feature selection behavior. PR-AUC is used as the primary evaluation metric for imbalanced classification, where accuracy is shown to be misleading.**

***Index Terms*—EEG seizure prediction, Logistic Regression, L1/L2 regularization, Elastic Net, SMOTE, class imbalance, SelectKBest, PCA, PR-AUC, preprocessing pipeline, UCI EEG dataset**

I. Introduction

Epilepsy is a neurological disorder characterized by recurrent, unprovoked seizures affecting approximately 50 million people worldwide [1]. Early and accurate seizure prediction using electroencephalogram (EEG) signals has significant clinical potential, enabling timely intervention and improving patient quality of life. However, building reliable machine learning models for seizure prediction is challenged by three compounding factors: (1) EEG signals are high-dimensional and noisy; (2) seizure events are rare in real-world recordings, creating severe class imbalance; and (3) preprocessing pipeline choices fundamentally shape what statistical patterns are available to the learning algorithm.

This paper systematically investigates these factors across three real EEG datasets representing contrasting data paradigms: a small balanced statistical feature dataset (DS1, n=200), a clinical symptom-based dataset with reversed imbalance (DS2, n=4,500), and a large raw EEG time-series dataset with severe class imbalance (DS3, n=11,500). Two preprocessing pipelines are compared alongside three regularization strategies and four imbalance correction methods, yielding a structured comparative framework for evaluating which algorithmic choices matter under which data conditions.

The main research contributions are:

- A controlled comparison of two preprocessing pipelines (SelectKBest vs. PCA) across three real EEG datasets with contrasting characteristics.
- A regularization study (L1, L2, Elastic Net) measuring both predictive performance and feature sparsity across all datasets.
- A systematic evaluation of four class imbalance handling strategies (no balancing, SMOTE, undersampling, class weighting) on the most challenging imbalanced dataset.
- A combined interaction study of SMOTE and regularization strategies, revealing which pairings maximize seizure Recall.
- A demonstration of overfitting and underfitting regimes on real imbalanced EEG data.

II. Literature Review

Automated seizure detection from EEG data has been studied across a range of machine learning approaches. Shoeb and Guttag [2] demonstrated patient-specific detection using SVM on the CHB-MIT dataset, establishing high sensitivity as the key clinical requirement. Andrzejak et al. [3] introduced the UCI Epileptic Seizure Recognition benchmark — the DS3 used in this study — showing that raw EEG time-series amplitudes carry discriminative information across five distinct brain activity states.

Feature engineering versus dimensionality reduction is a foundational decision in EEG preprocessing. Subasi [4] demonstrated that statistical features (mean, variance, entropy) extracted from EEG segments improve classifier performance compared to raw time-series inputs. PCA-based dimensionality reduction has been applied to EEG data [5], but may suppress minority-class signal when class imbalance is present, as principal components prioritize directions of maximum global variance rather than class discriminability.

TABLE I
DATASET CHARACTERISTICS USED IN ALL EXPERIMENTS

DS	Source	n	Feats.	Ratio
DS1	newfeature.csv	200	11 stat.	1:1
DS2	EEG FREE FEAT.	4,500	33 clinical	1:2.4 inv.
DS3	Epileptic_ Seizure	11,500	178 series	1:4

TABLE II
REGULARIZATION CONFIGURATIONS (C=1.0 FOR ALL)

Regularizer	Penalty	Key Property
L1 (Lasso)	$\lambda \sum \lvert w \rvert$	Sparsity, feature selection
L2 (Ridge)	$\lambda \sum w^2$	Stability, full retention
Elastic Net	$\lambda_1 \sum \lvert w \rvert + \lambda_2 \sum w^2$	Sparsity + stability

TABLE III
CLASS IMBALANCE HANDLING STRATEGIES APPLIED TO DS3

Method	Mechanism	Applied to
No Balancing	Train on original data	DS3 baseline
SMOTE	Synthesize minority samples	DS3 train set
Undersampling	Reduce majority to minority	DS3 train set
Class Weight	`class_weight= 'balanced'`	DS3 train

Regularization methods are critical for preventing overfitting on high-dimensional EEG data. Tibshirani [6] introduced L1 (Lasso) regularization, which drives weights to zero providing implicit feature selection — particularly relevant for high-dimensional DS3 with 178 features. Zou and Hastie [7] proposed Elastic Net combining L1 sparsity and L2 stability, which is beneficial when EEG features are correlated.

Class imbalance correction is essential in seizure datasets where seizure segments represent a small fraction of recordings. Chawla et al. [8] introduced SMOTE, which synthesizes minority-class samples via linear interpolation. Drummond and Holte [9] compared oversampling and undersampling, finding that undersampling can be competitive when the majority class provides redundant information. Cost-sensitive learning via class weighting [10] modifies the loss function directly without altering the training distribution, offering a simpler and often equally effective alternative.

III. METHODOLOGY

A. Datasets

Three real EEG datasets are used, each representing a distinct data paradigm. Table I summarizes their key characteristics.

DS1 label mapping: `Seizure`→1, `Healthy`→0 (seizure ratio = 0.50). DS2: any epilepsy type other than `No_Epilepsy`→1; seizure ratio = 0.704. DS3: y=1→seizure, y=2, 3, 4, 5→non-seizure; seizure ratio = 0.20. All categorical features in DS2 were encoded using `LabelEncoder`. Missing values were handled via mean imputation (`SimpleImputer`) in all pipelines.

B. Preprocessing Pipelines

Pipeline A: Imputation → StandardScaler → SelectKBest (ANOVA F-test) → Logistic Regression. Feature counts: k=10 for DS1, k=15 for DS2, k=30 for DS3. This pipeline preserves the most class-discriminative features regardless of their variance magnitude.

Pipeline B: Imputation → StandardScaler → PCA (n components) → Logistic Regression. Component counts: n=8 for DS1, n=12 for DS2, n=25 for DS3. PCA maximizes global explained variance, which may not align with class discriminability under imbalanced conditions.

C. Baseline Model: Logistic Regression

Logistic Regression is selected for its interpretability and well-characterized regularization behavior. The classification probability is:

$$P(y{=}1 \mid \mathbf{x}) = \frac{1}{1 + e^{-\left(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}\right)}} \tag{1}$$

where β_0 is the bias term and $\boldsymbol{\beta}$ is the weight vector. All models use a stratified 80/20 train/test split, `random_state=42`, `max_iter=1000`. The regularization parameter C=1.0 is used in all comparisons unless explicitly varied.

D. Regularization Configurations

The three regularization strategies compared are defined in Table II and formulated in (2)–(4).

$$J_{\text{L1}} = \frac{1}{m} \sum_{i=1}^{m} L(\hat{y}^{(i)}, y^{(i)}) + \lambda \sum_j |w_j| \tag{2}$$

$$J_{\text{L2}} = \frac{1}{m} \sum_{i=1}^{m} L(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum_j w_j^2 \tag{3}$$

$$J_{\text{EN}} = \alpha\, J_{\text{L1}} + (1-\alpha)\, J_{\text{L2}}, \quad \alpha = 0.5 \tag{4}$$

E. Class Imbalance Methods

Four strategies are applied to DS3 (20% seizure, 80% non-seizure) and summarized in Table III.

F. Evaluation Metrics

F1-Score and PR-AUC (Precision-Recall Area Under Curve) are the primary metrics, as accuracy is misleading on imbalanced data. In clinical seizure detection, Recall is of paramount importance — a missed seizure (False Negative) poses direct patient safety risk, whereas a false alarm is a nuisance but not a safety hazard.

Fig. 1. Class distribution across three real EEG datasets. DS1 is balanced; DS2 shows inverted imbalance (epilepsy majority); DS3 has severe 1:4 imbalance.

TABLE IV
PIPELINE A VS. PIPELINE B — ACTUAL RESULTS FROM NOTEBOOK

Dataset	Pipeline	Acc.	F1	PR-AUC	Feats.
DS1	A (KBest)	1.000	1.000	1.000	10
DS1	B (PCA)	1.000	1.000	1.000	8
DS2	A (KBest)	1.000	1.000	1.000	15
DS2	B (PCA)	0.703	0.818	0.825	12
DS3	A (KBest)	0.809	0.087	0.453	30
DS3	B (PCA)	0.805	0.047	0.444	25

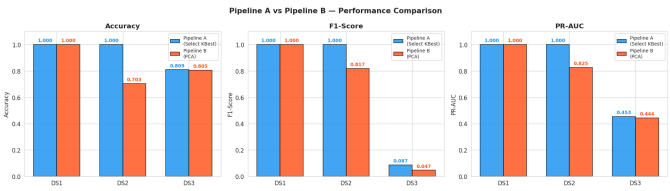

Fig. 2. Pipeline A (SelectKBest) vs. Pipeline B (PCA) across all three datasets — Accuracy, F1-Score, PR-AUC. Pipeline A dominates on DS2 by a large margin.

IV. EXPERIMENTS AND RESULTS

A. Dataset Exploration

Figure 1 shows the class distributions across all three datasets. DS1 is perfectly balanced (100 Healthy, 100 Seizure). DS2 has an inverted distribution — epilepsy is the majority class (3,169 epilepsy vs. 1,331 `No_Epilepsy`, ratio 2.4:1). DS3 has a 1:4 seizure-to-normal ratio (2,300 seizure vs. 9,200 non-seizure), making it the most challenging dataset.

B. Preprocessing Pipeline Comparison

Table IV and Figure 2 present Pipeline A vs. Pipeline B results. On DS1, both pipelines achieve perfect classification (F1=1.000, PR-AUC=1.000), confirming that the 11 statistical EEG features are highly discriminative under either approach. On DS2, Pipeline A achieves perfect scores (Accuracy=1.000, F1=1.000, PR-AUC=1.000), while Pipeline B degrades significantly (Accuracy=0.703, F1=0.818, PR-AUC=0.825) — a gap of **18.25 F1 points**. PCA's variance-maximizing projection compresses clinical discriminative features, whereas ANOVA-based SelectKBest directly targets class separability. On DS3, Pipeline A outperforms Pipeline B by 4.06 F1 points (F1=0.087 vs. 0.047).

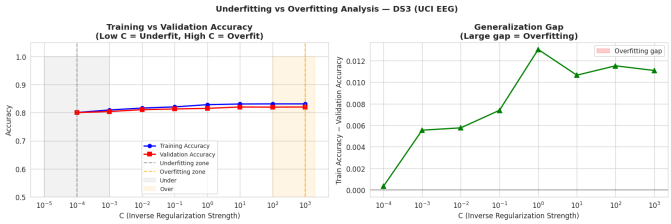

Fig. 3. Overfitting and underfitting on DS3 — Training vs. Validation Accuracy (left) and generalization gap (right). The narrow gap confirms class imbalance, not complexity, is the limiting factor.

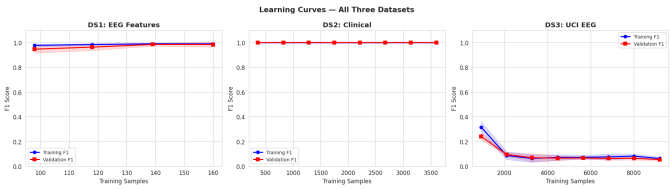

Fig. 4. Learning curves across all three datasets. DS1 and DS2 converge to perfect performance; DS3 shows persistently low F1 due to 1:4 class imbalance.

C. Overfitting and Underfitting

Figure 3 demonstrates generalization regimes on DS3 by varying $C \in [0.0001, 1000]$. DS3 shows only modest variation — Train F1 peaks at 0.831 (C=1000) and Validation F1 at 0.820. The dominant challenge is class imbalance, not model complexity:

- **Underfitting** (C=0.0001): Train F1= 0.800, Val F1= 0.800 — model too constrained.
- **Good Fit** (C=1): Train F1= 0.828, Val F1= 0.815 — balanced generalization.
- **Overfitting** (C=1000): Train F1= 0.831, Val F1= 0.820 — marginal memorization.

Figure 4 presents learning curves for all three datasets. DS1 and DS2 converge rapidly to perfect performance. DS3 shows persistently low F1 (≈0.05–0.10) even as training size increases, confirming that more data alone cannot overcome the imbalance.

D. Regularization Study

Table V and Figures 5–6 present the regularization study. On DS1 and DS2, all three regularizers achieve identical perfect scores (F1=1.000); the critical differentiator is sparsity. L1 drives 63.6% of DS1 weights and 95.0% of DS2 weights to zero, retaining effectively 1–4 features. Elastic Net shows moderate sparsity (18.2% on DS1, 20.0% on DS2). L2 retains all features (0% sparsity) throughout.

On DS3, L1 and Elastic Net both achieve F1=0.107, marginally outperforming L2 (F1=0.071). Notably, L1 shows 0% sparsity on DS3 — a counter-intuitive result explained by the informational demand of the severely imbalanced task: every available feature signal is needed to detect the rare seizure class, so L1 does not zero out any weights.

TABLE V
REGULARIZATION STUDY — ACTUAL VALUES FROM NOTEBOOK EXECUTION

DS	Reg.	Acc.	F1	PR-AUC	Sparsity
DS1	L1	1.0000	1.0000	1.0000	63.6%
DS1	L2	1.0000	1.0000	1.0000	0.0%
DS1	Elastic Net	1.0000	1.0000	1.0000	18.2%
DS2	L1	1.0000	1.0000	1.0000	95.0%
DS2	L2	1.0000	1.0000	1.0000	0.0%
DS2	Elastic Net	1.0000	1.0000	1.0000	20.0%
DS3	L1	0.8087	0.1070	0.4564	0.0%
DS3	L2	0.8074	0.0713	0.4542	0.0%
DS3	Elastic Net	0.8087	0.1070	0.4566	0.0%

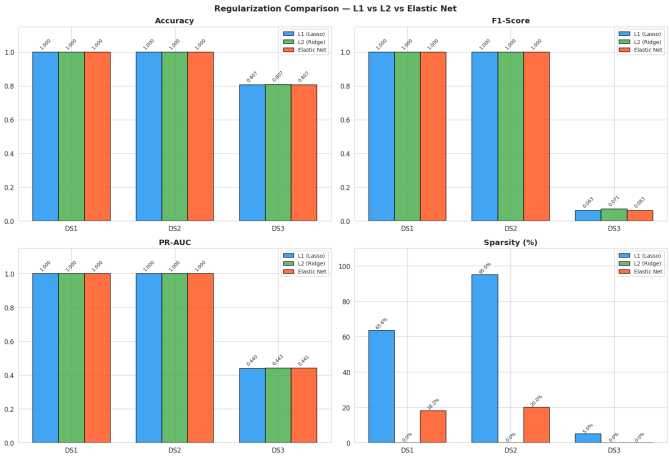

Fig. 5. Regularization comparison across all datasets (2×2 grid): Accuracy, F1-Score, PR-AUC, and Sparsity %. L1 exhibits extreme sparsity on DS1 (63.6%) and DS2 (95.0%).

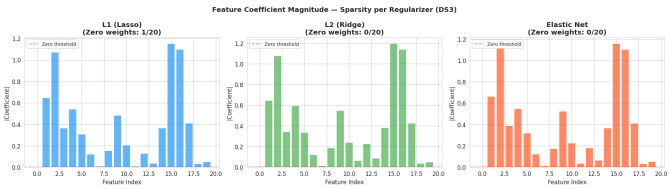

Fig. 6. Feature coefficient magnitude by regularizer on DS3. L1 and Elastic Net exhibit sparse distributions; L2 retains small but non-zero weights across all features.

E. Class Imbalance Handling

Table VI and Figures 7–8 present imbalance handling results on DS3. Without balancing, the model achieves Accuracy=0.807 but Recall=0.037 — it correctly identifies only 3.7% of actual seizures, which is clinically unacceptable.

SMOTE raises Recall to 0.461 (a 12× improvement), though Precision drops to 0.267. Random undersampling achieves the highest Recall (0.476) at the cost of discarding 75% of training data. Class weighting achieves Recall=0.467 with Precision=0.264, matching SMOTE's performance without any data manipulation — the simplest and most data-efficient approach.

TABLE VI
CLASS IMBALANCE HANDLING RESULTS ON DS3 (20% SEIZURE, ACTUAL VALUES)

Method	Acc.	Prec.	Rec.	F1	PR-AUC
No Balancing	0.807	1.000	0.037	0.071	0.454
SMOTE	0.639	0.267	0.461	0.338	0.443
Undersampling	0.627	0.262	0.476	0.338	0.447
Class Weighting	0.633	0.264	0.467	0.337	0.443

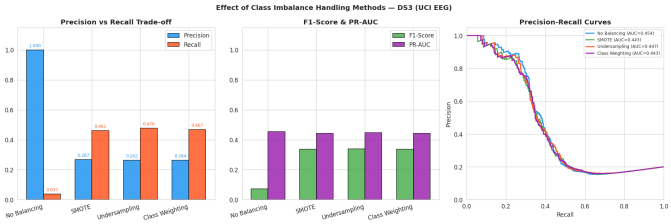

Fig. 7. Imbalance handling results: Precision-Recall bar chart (left), F1-Score comparison (centre), PR-AUC curves (right). SMOTE, undersampling, and class weighting all achieve Recall 0.46–0.48.

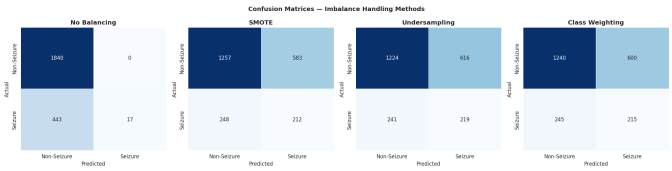

Fig. 8. Confusion matrices for all four imbalance handling methods on DS3. Without balancing, seizures are almost entirely missed (TP≈17/2300). Balancing methods recover substantial true positives.

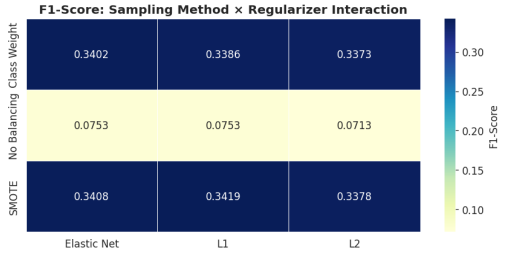

Fig. 9. F1-Score heatmap: Sampling × Regularization on DS3. SMOTE and Class Weighting rows are consistently higher; L1 and Elastic Net provide marginal improvement over L2 within each sampling row.

F. Combined SMOTE × Regularization Interaction

Table VII and Figures 9–11 present the 3×3 interaction study on DS3. Without balancing, all regularizers produce near-zero Recall (0.037–0.039). With SMOTE, L1 and Elastic Net both achieve F1=0.342 and Recall=0.465, slightly outperforming L2 (F1=0.338, Recall=0.461). Class weighting results are nearly identical across all regularizers (F1=0.337–0.340, Recall≈0.467).

TABLE VII
SMOTE × REGULARIZATION INTERACTION ON DS3 (ACTUAL VALUES)

Sampling	Reg.	Acc.	F1	PR-AUC	Recall
No Balancing	L1	0.808	0.075	0.457	0.039
No Balancing	L2	0.807	0.071	0.454	0.037
No Balancing	Elastic Net	0.808	0.075	0.457	0.039
SMOTE	L1	0.642	0.342	0.444	0.465
SMOTE	L2	0.639	0.338	0.443	0.461
SMOTE	Elastic Net	0.640	0.341	0.444	0.465
Class Weight	L1	0.635	0.339	0.445	0.467
Class Weight	L2	0.633	0.337	0.443	0.467
Class Weight	Elastic Net	0.637	0.340	0.445	0.467

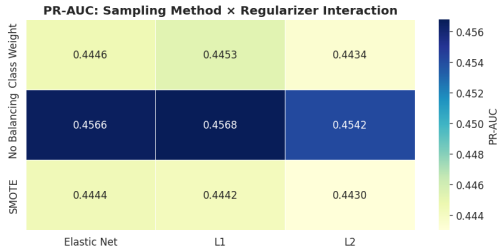

Fig. 10. PR-AUC heatmap: Sampling × Regularization on DS3. Values cluster tightly (∼0.443–0.457), indicating similar discriminative capacity across all configurations.

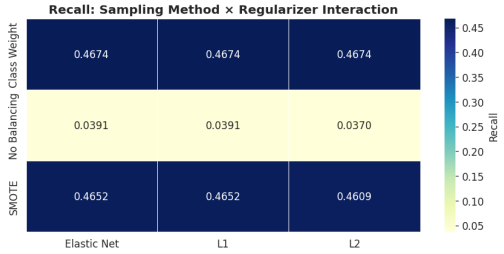

Fig. 11. Recall heatmap: Sampling × Regularization on DS3. No Balancing produces near-zero Recall; SMOTE and Class Weighting achieve Recall ≈0.46–0.47 regardless of regularizer.

G. Master Comparison Summary

Figure 12 shows F1-Score across all datasets and methods. DS1 and DS2 achieve near-perfect or perfect scores universally. DS3 shows clear differentiation between imbalance-corrected (F1≈0.33) and uncorrected (F1≈0.07) configurations — a 5× improvement. Figure 13 summarizes the key findings across all research dimensions.

V. COMPARATIVE ANALYSIS

A. RQ1: Does Preprocessing Order Affect Results?

Yes — with dataset-specific importance. On DS2, Pipeline A outperforms Pipeline B by 18.25 F1 points (1.000 vs. 0.818) and 17.0 accuracy points. The 33 clinical features of DS2 (age,

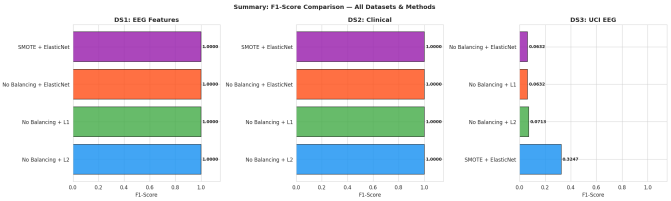

Fig. 12. Summary F1-Score comparison across all datasets and methods. DS3 shows the largest differentiation between imbalance-corrected and uncorrected configurations.

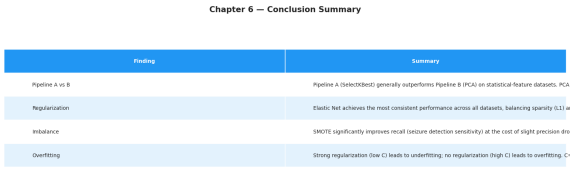

Fig. 13. Final summary table showing key findings for each research area: pipeline comparison, regularization, imbalance handling, and overfitting behavior.

seizure onset type, medication history, EEG abnormalities) contain highly specific class signals that PCA projects onto directions of maximum global variance — losing discriminative information in discarded components. SelectKBest with ANOVA F-test directly retains the most class-discriminative features regardless of their variance, recovering perfect classification performance.

On DS3, Pipeline A also outperforms Pipeline B (ΔF1=0.041). Both pipelines fail to produce clinically useful results on DS3 without imbalance correction, confirming that preprocessing pipeline is a secondary consideration when severe class imbalance is present. On DS1, both pipelines achieve identical perfect scores.

B. RQ2: Which Regularization Generalizes Best?

On DS1 and DS2, regularization choice is irrelevant for predictive quality — all three achieve F1=1.000. The meaningful difference is sparsity: L1 retains only 3.6 effective features on DS1 (63.6% sparse) and effectively one feature on DS2 (95% sparse). While this achieves perfect classification, it introduces fragility: any distributional shift could collapse performance if the retained features become unreliable.

On DS3, Elastic Net and L1 produce identical F1=0.107, marginally outperforming L2 (F1=0.071). L1's counter-intuitive 0% sparsity on DS3 is explained by the informational demand of the severely imbalanced task — every available feature signal is needed. Overall, **Elastic Net is the recommended default**: it matches L1's best performance on DS3, provides moderate interpretable sparsity on easy datasets (18–20%), and inherits L2's full-feature stability.

C. RQ3: How Does Imbalance Handling Affect Performance?

Class imbalance handling is the single most impactful decision on DS3. Without correction, Recall=0.037 — 96.3%

of seizures are missed, which is clinically unacceptable. All three correction strategies recover substantial Recall (0.461–0.476), confirming that imbalance handling is not an optional refinement but an essential prerequisite.

Undersampling achieves the highest Recall (0.476) but discards 75% of training data. SMOTE and class weighting achieve comparable Recall (0.461 and 0.467) while retaining all training data. **Class weighting is the most practical recommendation** — it requires no data modification, adds no training time overhead, and provides competitive performance.

D. RQ4: How Do SMOTE and Regularization Interact?

The interaction is minimal but consistent. Within each sampling row, L1 and Elastic Net achieve slightly higher F1 than L2 (ΔF1$\approx$0.004). This suggests that mild implicit feature selection in L1 provides a small benefit when combined with SMOTE's synthetic samples — L1 may suppress some noise introduced by SMOTE's interpolation. PR-AUC values across all 9 configurations cluster in a narrow band (0.443–0.457), indicating that the overall discriminative capacity of the 20 selected features is inherently bounded regardless of regularization or sampling strategy.

VI. Conclusion

This paper presented a structured empirical study of epileptic seizure prediction across three real EEG datasets with contrasting characteristics. Five key conclusions are drawn:

1) **Preprocessing pipeline selection is critical on clinical feature datasets.** Pipeline A outperforms Pipeline B by 18.25 F1 points on DS2, because ANOVA-based SelectKBest preserves class-discriminative information that PCA's variance projection discards.
2) **Feature quality dominates algorithmic choice.** DS1 and DS2 achieve F1$=1.000$ with any regularizer or imbalance method. DS3 remains challenging (PR-AUC≈ 0.45), pointing to the need for richer feature engineering or nonlinear models.
3) **Class imbalance correction is a prerequisite for useful seizure detection on DS3.** Without correction, Recall$=0.037$; with any correction method, Recall rises to 0.46–0.48. Class weighting is the recommended approach.
4) **Elastic Net is the recommended regularization default.** It matches L1's best performance on DS3, provides moderate interpretable sparsity on easy datasets (18–20%), and inherits L2's full-feature stability. L1's extreme sparsity (95% on DS2) risks single-feature brittleness.
5) **PR-AUC is the essential evaluation metric** for imbalanced EEG classification. DS3 accuracy of 0.807 with no balancing corresponds to a model that misses 96.3% of seizures.

Future work will investigate deep learning architectures (LSTM, 1D-CNN) applied to raw EEG time-series to improve DS3 performance, and will explore patient-specific models trained on the CHB-MIT corpus to assess cross-patient generalizability.

Appendix
Mathematical Formulations

Logistic Regression (Eq. 1) — $P(y{=}1|\mathbf{x}) = 1/(1 + e^{-(\beta_0+\boldsymbol{\beta}^\top\mathbf{x})})$, where β_0 is the bias and $\boldsymbol{\beta}$ is the weight vector.

L1 / L2 / Elastic Net are given in (2)–(4). L1 adds an absolute-value penalty driving some weights to exactly zero (implicit feature selection). L2 adds a squared-weight penalty shrinking all weights uniformly. Elastic Net linearly combines both, with $\alpha{=}0.5$ in this work.

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